# Response to Reviewer #4

Ms. Ref. No.: MEAS-D-26-10707

Gradient-Sensitive Functional Descriptors for Interferometric Surface Metrology: Complementing Amplitude-Averaging ISO Parameters

*Measurement* — Revised version, September 2026

The demanding and detailed review of Reviewer #4 is gratefully acknowledged. The four points call, in their fullest form, for a new experimental programme — independent runs, thermal cycles, re-centring, a complete GUM Type-B budget, physical defects, additional materials and diameters, and an inter-laboratory comparison. This is a large scope, and not all of it can be executed within a single revision cycle. The reviewer's own alternative instructions have therefore been followed: what the existing data genuinely support has been done (a quantified type-B budget, a defect-to-background scaling analysis, and a standardised sampling/angular-unit treatment), and, for what requires new experiments, the claims have been sharply narrowed and a concrete validation programme defined rather than overstating the evidence. Each point is addressed below; a marked (redline) version of the manuscript accompanies this letter.

## Point 4.1 — Between-run reproducibility and a GUM-compliant uncertainty budget

> *Expand the uncertainty analysis significantly throughout the manuscript. The current version relies solely on within run repeatability derived from five continuous sweeps [...]. To address this major weakness you must provide a comprehensive between run reproducibility assessment. You are specifically required to include new experimental data involving completely independent measurement setups distinct thermal cycles and physical recentering of the measurement object. Furthermore the manuscript currently lacks a full GUM compliant uncertainty budget [...]. These additions must include an assessment of laser wavelength stability refractive index variations Abbe error camera nonlinearity and form removal uncertainty.*

**Response:** The uncertainty analysis has been expanded as far as the existing data permit, and the manuscript is explicit about what it cannot yet establish. (1) A new subsection (Sec. 3.8, "Indicative Type-B Uncertainty Budget", with new Table 2) quantifies exactly the five contributions named: *laser wavelength stability* (±1 nm module tolerance, rectangular: 0.11% relative, 0.3 nm on the FN 111 *RONt*; a pure scale factor that cannot create or mask localised features); *refractive-index variation of air* ($< 10^{-5}$ relative at the observed ±0.2°C stability: negligible); *Abbe error / axis wobble* (±12 $\mu$rad, analysed through both the cosine-error and beam-walk mechanisms: sub-0.01 nm after LSCI first-harmonic removal); *camera nonlinearity / EFM localisation* (bounded empirically by the ±0.05-fringe per-frame uncertainty, reduced by band-limited averaging to $\approx 1.1$ nm); and *form-removal uncertainty* (limaçon second-order term $a^2/2R$, evaluated at 7–41 nm for centring eccentricities of 15–35 $\mu$m, carried as a bounded systematic because it is common to all sweeps, and consistent with the 34 nm optical–tactile *RONt* difference that the $E_n = 0.61$ comparison shows to be within combined uncertainties). The combined random budget confirms the type-A term dominates and identifies the form-removal systematic and the untested between-run effects as the leading unquantified risks. (2) **New experimental data have been added in this revision:** an independent static stability session (24,000 frames over 4,000 s — the same duration as the FN 111 acquisition — recorded in a separate session with the object not rotating) provides a direct measured bound on instrument drift and the phase noise floor under realistic laboratory conditions: end-to-end drift −82 nm ($-1.1\ \mathrm{nm\,min^{-1}}$, $\approx 15$ nm per 800 s sweep-equivalent), detrended residual 9.4 nm SD (45 nm peak-to-valley), and a frame-to-frame noise floor of 2.1 nm (0.008 fringe), confirming the ±0.05-fringe EFM bound as conservative (Sec. 3.8, Table 2). (3) Regarding between-run reproducibility of the *rotating* measurement with independent setups, thermal cycles and physical re-centring: these are agreed to be required before the descriptors can be recommended for adoption, but they cannot be performed within this revision cycle, as the interferometer is no longer in the exact configuration used for the archives. Rather than present hastily acquired data, this limitation is stated explicitly and in the reviewer's own terms. The paper's claims are scoped throughout as proof-of-concept and within-run only, and they are believed to be commensurate with the evidence presented.

**Changes made:** New Sec. 3.8 and Table 2, including a measured drift/noise-floor bound from an independent static stability session; Sec. 3.7 renamed "Uncertainty Evaluation" with a pointer to the type-B budget and the GUM reference (JCGM 100); Abstract, Limitations (first point), Data Availability (static recording available on request) and Future Work item (i) updated. No claim is made that five continuous sweeps provide full reproducibility; the text repeatedly states these are within-run lower bounds.

## Point 4.2 — Synthetic defects and translation to real-world conditions

> *Improve and broaden the synthetic defect injection methodology to reflect real world conditions. [...] If physical testing is absolutely impossible you need to substantially expand the discussion section. You must explain exactly how your synthetic sensitivity results can be reliably translated to practical industrial inspection scenarios such as the rigorous quality control of bearing surfaces complex optical components or precision mechanical seals.*

**Response:** Calibrated physical-defect artefacts are not available within the revision period, so the reviewer's alternative instruction has been followed and the Discussion substantially expanded with a quantitative translation framework. A new paragraph in Sec. 5.3 derives a scaling rule: a localised defect adds a total-variation increment $\Delta TV$ that depends only on the defect geometry ($\Delta TV \approx 2A$ for a spike of amplitude $A$), so the relative full-profile BV-density response is $\Delta TV/TV_{\text{background}}$. This reproduces the observed +18% on the FN 111 ($TV = 1.137$ $\mu$m) and predicts only +0.17% for the identical spike on the FN 101 ($TV = 118.4$ $\mu$m) — a direct quantification of the reviewer's observation that the FN 101 defect-to-background ratio is roughly 500 times less favourable. Two practical consequences are drawn: full-profile BV density is best suited to smooth finished surfaces (bearing raceways, optical components, precision seals) whose residual amplitudes are comparable to the FN 111 case, with the synthetic archetypes mapped to realistic feature classes (spike $\leftrightarrow$ pit/inclusion edge, scratch $\leftrightarrow$ tooling or handling mark, sinusoid $\leftrightarrow$ chatter); and for rougher backgrounds the sliding-window formulation restores local contrast, which is exactly why the Flick flat is detected on the FN 101 despite the unfavourable full-profile ratio. The text states plainly that validation against physical defects of calibrated dimensions remains necessary before these scaling arguments can be considered established.

**Changes made:** New paragraph in Sec. 5.3; Limitations third point rewritten to reference the scaling analysis; Future Work item (iii) upgraded to calibrated physical defects and a blind-detection protocol.

## Point 4.3 — Sampling-interval dependence and angular-unit conventions

> *Address the complex sampling interval dependence of the newly proposed functional descriptors. [...] they require a rigorous and mathematically sound conversion to angular units such as nanometers per degree for any meaningful cross instrument*

*comparison. [...] You must provide a clear standardized procedure [...]. The development and proposal of agreed angular unit conventions is absolutely critical at this stage.*

**Response:** This is agreed to be the most technically important point, and it has been turned into an explicit element of the methodology (new Sec. 3.6, "Angular-Unit Convention for Sampling-Dependent Descriptors"). The convention has three steps: (1) *band-limit first* — apply a declared phase-correct Gaussian profile filter (ISO 16610-21) at a stated UPR cutoff, after which the profile derivative is bounded and the discrete sums converge to the continuous integrals as $\Delta\alpha \to 0$, so the descriptor characterises the declared bandwidth rather than the sampling grid; (2) *sample adequately* — at least ten samples per cutoff undulation (the present data carry ≈290 and ≈1200); (3) *report in angular and arc-length units together with the filter cutoff.* Crucially, the dimensional distinction the reviewer's comment points to is now made explicitly: amplitude descriptors ($R_a$, $R_q$, $L^1$, $L^2$, $RONt$) have the dimension of length (nm) and are sampling-independent, whereas the gradient descriptors are height-per-length and must be reported as $\mathrm{nm\,deg^{-1}}$ and, for diameter-independent cross-instrument comparison, as $\mathrm{nm\,mm^{-1}}$ of arc length ($\mathrm{BV}_s = TV/\pi D$). Converted values are given for both artefacts (FN 111: BV 3.2 $\mathrm{nm\,deg^{-1}}$ = 12.1 $\mathrm{nm\,mm^{-1}}$; FN 101: BV 329 $\mathrm{nm\,deg^{-1}}$ = 1.26 $\mu\mathrm{m\,mm^{-1}}$, at the 15-UPR cutoff), and the text notes that the nearly identical circumferences ($\approx$ 94 mm) make the large arc-length difference a genuine surface effect rather than a unit artefact. Because each conversion is a fixed multiplicative factor, relative uncertainties and relative defect responses are identical in every unit system. The parallel with declaring the filter for $RONt$ is drawn explicitly as the route to reference-software adoption; cross-instrument empirical verification is listed as Future Work.

**Changes made:** New Sec. 3.6 (with the arc-length $\mathrm{nm\,mm^{-1}}$ convention and the amplitude-versus-gradient dimensional distinction); Sec. 3.5 closing paragraph updated to point to it; *Practical considerations* (Sec. 5.3), Limitations (fourth point), Metrological Implications and Future Work updated to reference the convention.

## Point 4.4 — Validation scope and generalizability

> *Broaden the validation scope and definitively demonstrate the generalizability of your findings. The current experimental validation is far too narrow [...] strictly limited to merely two specific artifacts from a single commercial manufacturer JENOPTIK. [...] You need to rigorously validate your proposed methodology on a significantly wider range of artifact types [...]. Finally conducting detailed inter laboratory comparisons using a common artifact set would provide the vital long term reproducibility data [...].*

**Response:** It is agreed that generalisability is not yet demonstrated, and the revision now says so without qualification. New certified artefacts of different diameters and materials (optical glass, tungsten carbide) and an inter-laboratory campaign are beyond what can be added in this revision cycle; they require procurement and partner laboratories. The revision therefore (a) scopes every conclusion explicitly as a proof-of-concept on two artefacts; (b) notes the breadth the existing data do provide — the two artefacts differ substantially in material ($Al_2O_3$ ceramic versus steel) and by two orders of magnitude in residual amplitude, though they share diameter and manufacturer; and (c) converts the reviewer's requirements into a concrete validation programme in Future Work: artefacts of different diameters, materials and finishes; calibrated physical defects and blind detection; an inter-laboratory comparison with a common artefact set and standardised pipeline as the source of long-term reproducibility data for potential standardisation; and cross-instrument verification of the angular-unit convention. The revised claims are now strictly commensurate with the validation scope.

**Changes made:** Limitations second point rewritten; Future Work items (ii)–(v) expanded; proof-of-concept scoping verified throughout (Abstract, Sec. 5.3, Sec. 5.5).

It is hoped that the reviewer finds that the revision responds substantively to every point and that the claims are now properly bounded by the evidence.

The revised manuscript and the marked (redline) version accompany this response.